

5 Daemons

Performed by *Nikita Niakhai*

? Please, do not downgrade my work so much ;)

I don't like to talk about excuses also I know the holy grail of *the deadline*. Nevertheless, I will write a few words..

I had official excuse till 9th of October, because I was obliged to attend military commission in my country and then I spent two days travelling back to Innopolis, then I moved in dormitory again. And today (15.10) I am finally back to work.

P.S. As a result of my trip, I have found out, that **I am not fit for military service**, so I can freely proceed my master's degree!

▼ Task 1: Theory

1. What is `systemctl` and `init` and `systemd` ?

Learned `systemctl` via man → Learn `systemd`

systemctl - is a tool to control and introspect system manager and systemd.

systemd - is a service manager and core system that has a big ecosystem.
Main functions:

- * managing services and processes (start/end)
- * parallel booting of services
- * logging for services (via separate tool `journalctl`)
- * provides an interface to manage `cgroups` (- is a feature of Linux that gives

ability to create groups for processes in order to manage their resources/ set resources-limit and etc. / security limits)

* targets (= replacement for runlevels) - feature that divides process groups into main categories of user modes (text/ GUI/ networked), shutting down / rebooting modes

* timers (replacement for crontabs) - allows to create timers for processes in more integrated and powerful way in form of files. Inside folder

`etc/systemd/system` create 2 files `name.service` ([Unit] description of periodical process + [Service] path to executable service/process)+ `name.timer` ([Unit] description with time parameters, [Timer] commands to set the timer, [Install] configurational link for system to start the timer)

* mounting devices, swapping files, managing sockets, managing network connections/interfaces, managing user sessions

init - is a systemd that is run with PID=1 (process ID). it runs system, start essential daemons, sets up env and also shuts whole system down

2. what are the available Runlevel on linux?

Runlevel - feature of systemd that divides process groups into main categories of user modes (text/ GUI/ networked), shutting down / rebooting modes

0	System halt i.e., the system can be safely powered off with no activity.
1	Single user mode.
2	Multiple user mode with no NFS (network file system).
3	Multiple user modes under the command line interface and not under the graphical user interface.
4	User-definable.
5	Multiple user mode under GUI (graphical user interface) and this is the standard runlevel for most of the LINUX-based systems.
6	Reboot which is used to restart the system.

3. what does the systemctl list-unit-files command does?

The `systemctl list-unit-files` command shows all unit files that systemd knows about on the system in current memory. E.g., these are the configuration files that define services, timers and other units.

For each unit, it displays its **status (enabled/disabled)**, meaning whether it is configured to start automatically at boot or not, without actually showing whether the service is currently running. This allows to quickly manage units.

```
(nikita@nikita)-[~]
$ systemctl list-unit-files
```

UNIT FILE	STATE	PRESET
proc-sys-fs-binfmt_misc.automount	static	-
-.mount	generated	-
boot.mount	generated	-
dev-hugepages.mount	static	-
dev-mqueue.mount	static	-
media-cdrom0.mount	generated	-
proc-fs-nfsd.mount	static	-
proc-sys-fs-binfmt_misc.mount	disabled	disabled
run-lock.mount	disabled	enabled
run-rpc_pipefs.mount	generated	-
sys-fs-fuse-connections.mount	static	-
sys-kernel-config.mount	static	-
sys-kernel-debug.mount	static	-
sys-kernel-tracing.mount	static	-
tmp.mount	static	-
systemd-ask-password-console.path	static	-
systemd-ask-password-plymouth.path	static	-
systemd-ask-password-wall.path	static	-
session-2.scope	transient	-
accounts-daemon.service	enabled	enabled
apache-htcacheclean.service	disabled	disabled

▼ Task 2: Creating Systemd service

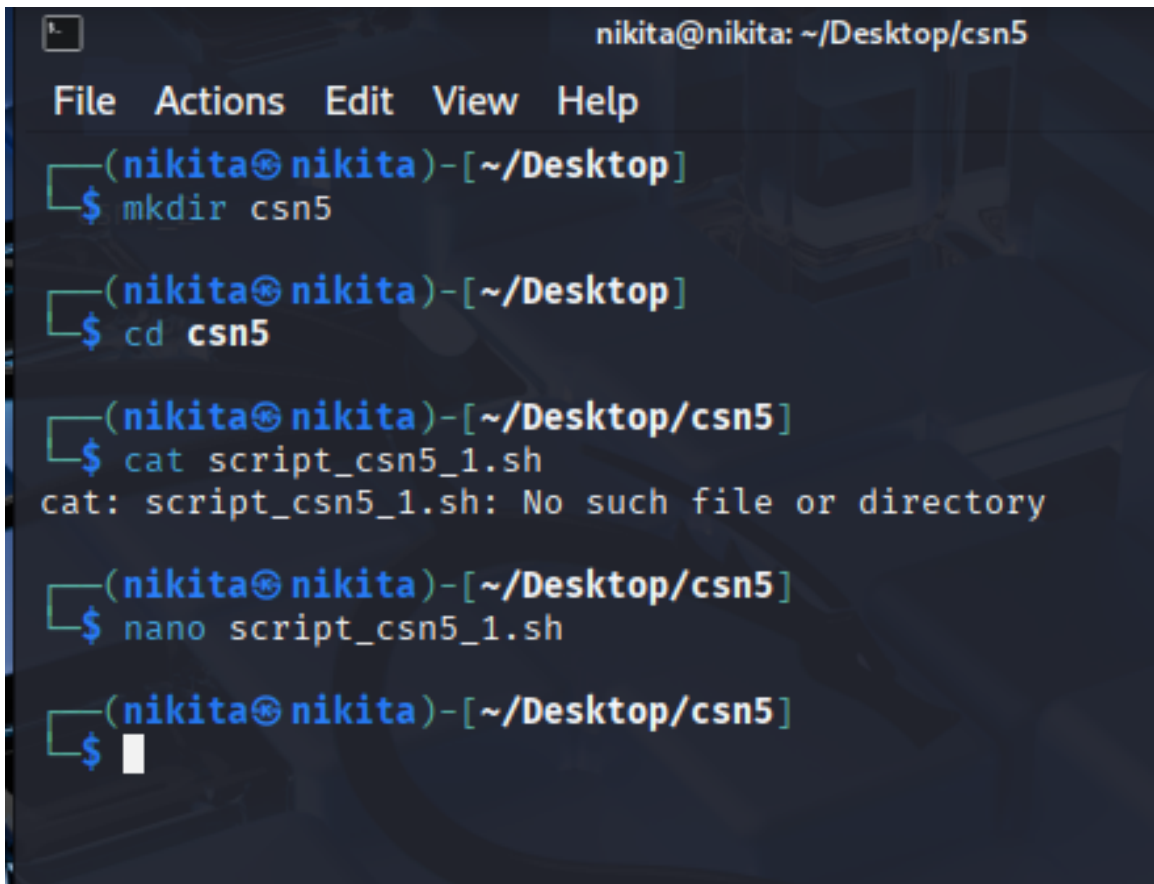
Create a shell script that would write a text to a file `echo "bla bla" > /tmp/test.txt` and then create a Systemd service that would this script. Here are the requirements:

1. It should run as your current user~
2. the working directory should be set to the current user home directory
3. this service will only run after the network.target service

I used for creating a script this source

[\[https://www.linuxjournal.com/content/how-create-shell-script-linux\]](https://www.linuxjournal.com/content/how-create-shell-script-linux)

Scripts for Linux are created inside files with `.sh` extension.

A terminal window with a dark background and light-colored text. The window title is 'nikita@nikita: ~/Desktop/csn5'. The menu bar shows 'File', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows the following commands and output:

```
(nikita@nikita)-[~/Desktop]
$ mkdir csn5

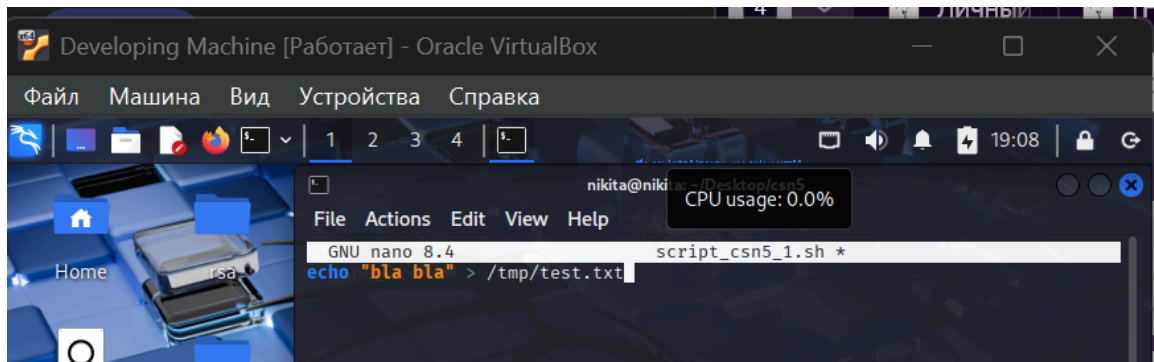
(nikita@nikita)-[~/Desktop]
$ cd csn5

(nikita@nikita)-[~/Desktop/csn5]
$ cat script_csn5_1.sh
cat: script_csn5_1.sh: No such file or directory

(nikita@nikita)-[~/Desktop/csn5]
$ nano script_csn5_1.sh

(nikita@nikita)-[~/Desktop/csn5]
$
```

I have created folder for this task on my Desktop and left a file with the script there.



I wrote script inside `.sh` file

```
(nikita@nikita)-[~/Desktop/csn5]
$ ./script_csn5_1.sh
zsh: permission denied: ./script_csn5_1.sh

(nikita@nikita)-[~/Desktop/csn5]
$ sudo chown nikita:nikita ./script_csn5_1.sh
[sudo] password for nikita:

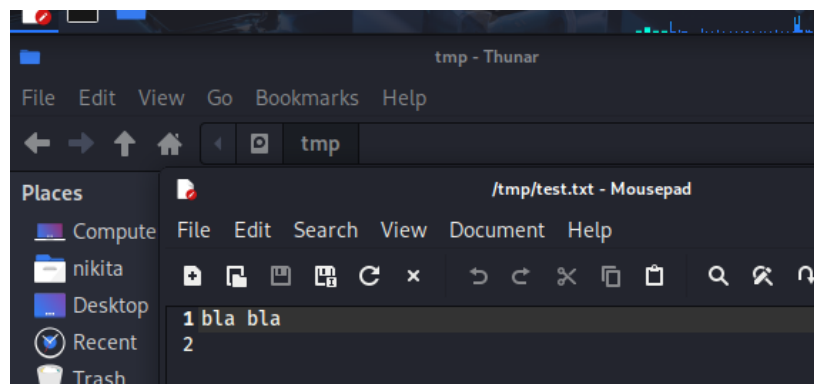
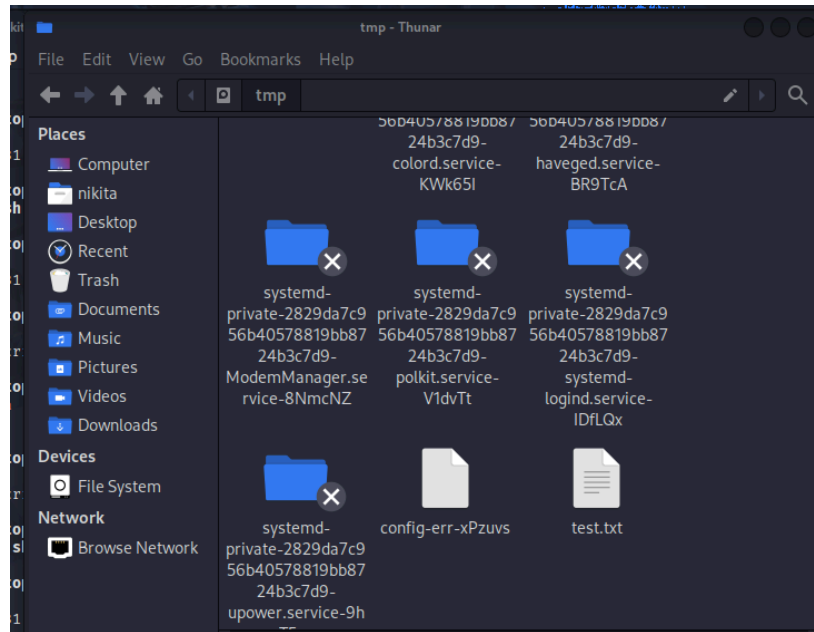
(nikita@nikita)-[~/Desktop/csn5]
$ ./script_csn5_1.sh
zsh: permission denied: ./script_csn5_1.sh

(nikita@nikita)-[~/Desktop/csn5]
$ chmod 777 script_csn5_1.sh

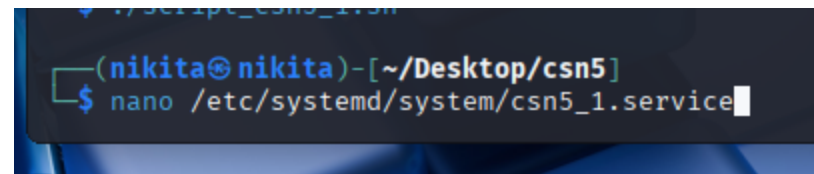
(nikita@nikita)-[~/Desktop/csn5]
$ ls -l script_csn5_1.sh
-rwxrwxrwx 1 nikita nikita 31 Oct 15 19:09 script_csn5_1.sh

(nikita@nikita)-[~/Desktop/csn5]
$ ./script_csn5_1.sh
```

For the file to be executable I need to change permissions.



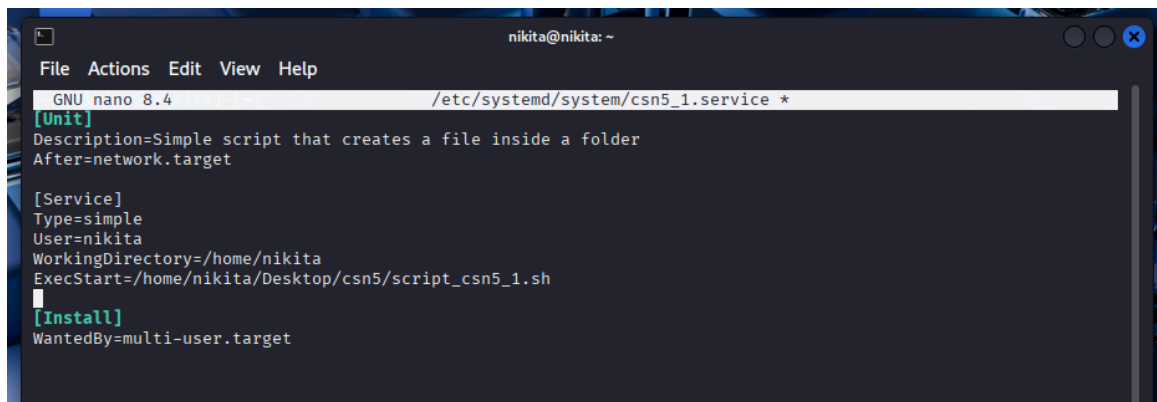
After successfully executing the script I saw the new file inside `tmp` folder



Creating a service

I reload daemon and enable (just once) service and then start it. But I got error, because path to ExecStart had `.` symbol. Now I fixed it

```
as a bad unit file setting.)
csn5_1.service:9: Neither a valid executable name nor an absolute path: ./home/nikita/Desktop/csn5/script_csn>
configuration has fatal error, unit will not be started.
~
```

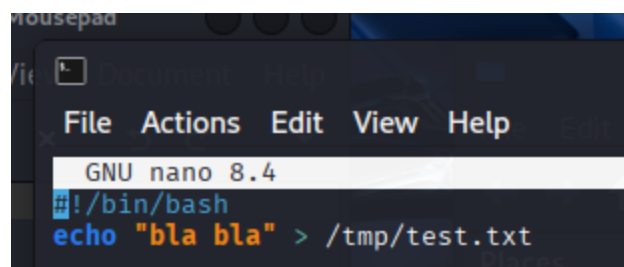


```
nikita@nikita: ~
File Actions Edit View Help
GNU nano 8.4 /etc/systemd/system/csn5_1.service *
[Unit]
Description=Simple script that creates a file inside a folder
After=network.target

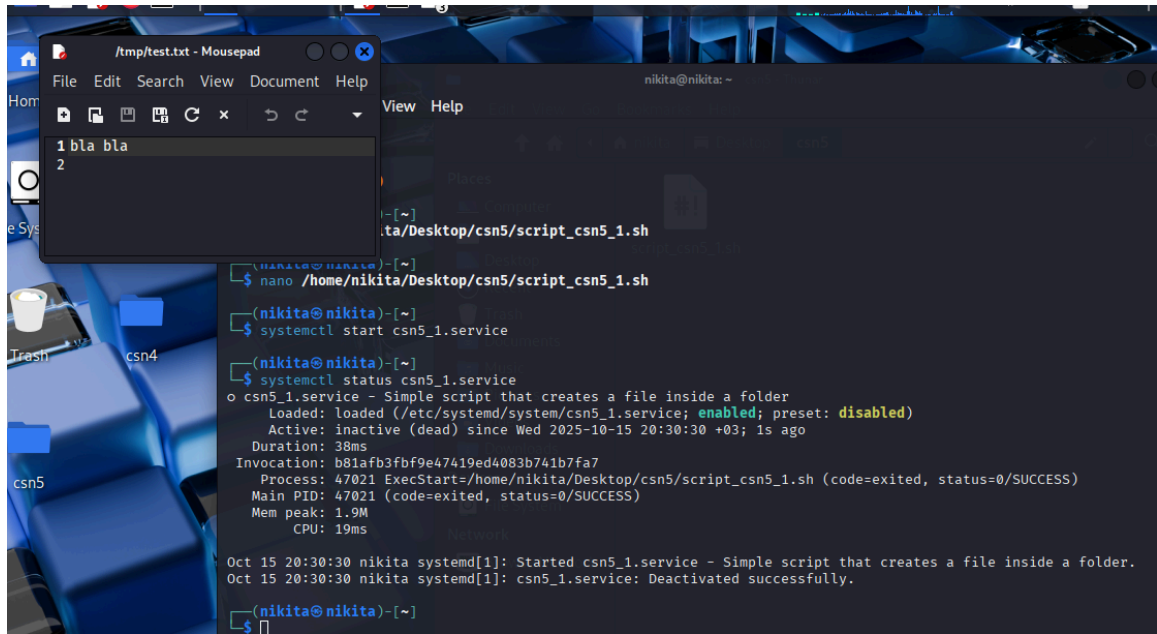
[Service]
Type=simple
User=nikita
WorkingDirectory=/home/nikita
ExecStart=/home/nikita/Desktop/csn5/script_csn5_1.sh

[Install]
WantedBy=multi-user.target
```

But then I got error 203 that file is not executable. Then I changed .sh file: add header (**first screenshot**) inside file. Everything works (**second screenshot**).



```
housepad
File Actions Edit View Help
GNU nano 8.4
#!/bin/bash
echo "bla bla" > /tmp/test.txt
```



```
nikita@nikita: ~  
$ nano /home/nikita/Desktop/csn5/script_csn5_1.sh  
$ systemctl start csn5_1.service  
$ systemctl status csn5_1.service  
o csn5_1.service - Simple script that creates a file inside a folder  
   Loaded: loaded (/etc/systemd/system/csn5_1.service; enabled; preset: disabled)  
   Active: inactive (dead) since Wed 2025-10-15 20:30:30 +03; 1s ago  
     Duration: 38ms  
  Invocation: b81afb3fbf9e47419ed4083b741b7fa7  
    Process: 47021 ExecStart=/home/nikita/Desktop/csn5/script_csn5_1.sh (code=exited, status=0/SUCCESS)  
   Main PID: 47021 (code=exited, status=0/SUCCESS)  
    Mem peak: 1.9M  
      CPU: 19ms  
  
Oct 15 20:30:30 nikita systemd[1]: Started csn5_1.service - Simple script that creates a file inside a folder.  
Oct 15 20:30:30 nikita systemd[1]: csn5_1.service: Deactivated successfully.  
$
```

Successfully runned a script (console pipeline + result of the file).

▼ Task 3: Install a web-server service

1. Update modules and dependencies `sudo apt update`
2. Check presence of nginx module

```

(nikita@nikita)-[~]
$ sudo apt update
[sudo] password for nikita:
Get:1 http://mirror.nl.cdn-perfprod.com/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirror.nl.cdn-perfprod.com/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://mirror.nl.cdn-perfprod.com/kali kali-rolling/main amd64 Contents (deb) [51.7 MB]
Get:4 http://mirror.nl.cdn-perfprod.com/kali kali-rolling/contrib amd64 Packages [114 kB]
Get:5 http://mirror.nl.cdn-perfprod.com/kali kali-rolling/contrib amd64 Contents (deb) [258 kB]
Get:6 http://mirror.nl.cdn-perfprod.com/kali kali-rolling/non-free amd64 Packages [187 kB]
Get:7 http://mirror.nl.cdn-perfprod.com/kali kali-rolling/non-free amd64 Contents (deb) [891 kB]
Fetched 74.1 MB in 3min 3s (404 kB/s)
1323 packages can be upgraded. Run 'apt list --upgradable' to see them.

(nikita@nikita)-[~]
$ nginx -v
nginx version: nginx/1.26.3

(nikita@nikita)-[~]
$ sudo systemctl status nginx
o nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:nginx(8)

```

3. Check nginx service status: is disabled

```

Docs: man:nginx(8)

(nikita@nikita)-[~]network
$ sudo useradd --system --no-create-home --shell /usr/sbin/nologin webuser

(nikita@nikita)-[~]

```

4. For privacy matters I will create low-privileged user

- Has no shell access
- Cannot log in
- Cannot modify system files

```
File Actions Edit View Help
GNU nano 8.4 /etc/systemd/system/secure-nginx.service *
[Unit]
Description=Secure Nginx Web Server
After=network.target

[Service]
Type=simple

# Run as restricted user
User=webuser
Group=webuser

# Use a safe working directory
WorkingDirectory=/var/www/html

# Command to start nginx directly
ExecStart=/usr/sbin/nginx -g 'daemon off;'
ExecReload=/bin/kill -s HUP $MAINPID
ExecStop=/bin/kill -s QUIT $MAINPID
Restart=on-failure
RestartSec=5s

# SECURITY HARDENING OPTIONS
NoNewPrivileges=true
PrivateTmp=true
ProtectSystem=strict
ProtectHome=true
ReadOnlyPaths=/usr /etc
ReadWritePaths=/var/log/nginx /var/www/html
ProtectKernelModules=true
ProtectKernelTunables=true
ProtectControlGroups=true
RestrictSUIDSGID=true
LockPersonality=true
MemoryDenyWriteExecute=true
CapabilityBoundingSet=CAP_NET_BIND_SERVICE
SystemCallFilter=@system-service
RestrictRealtime=true

# Logging
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target
```

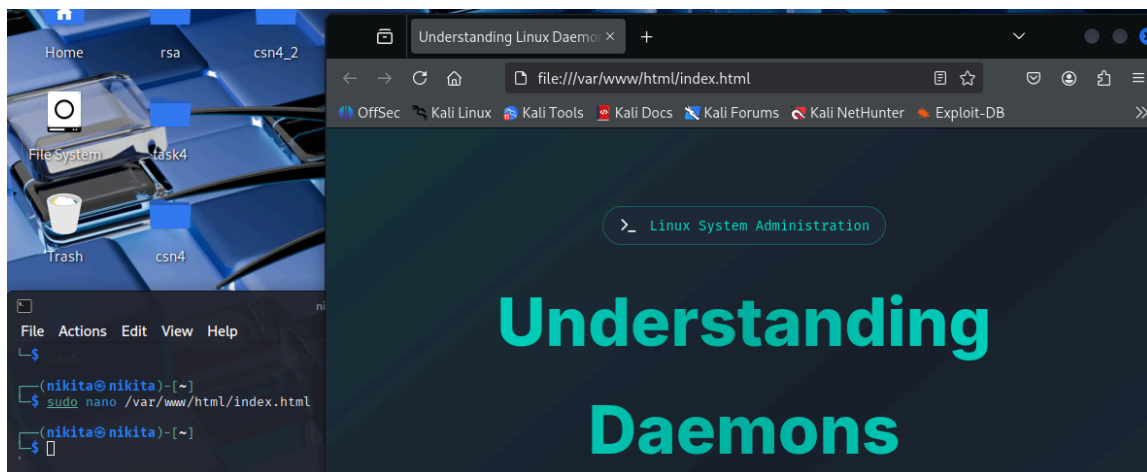
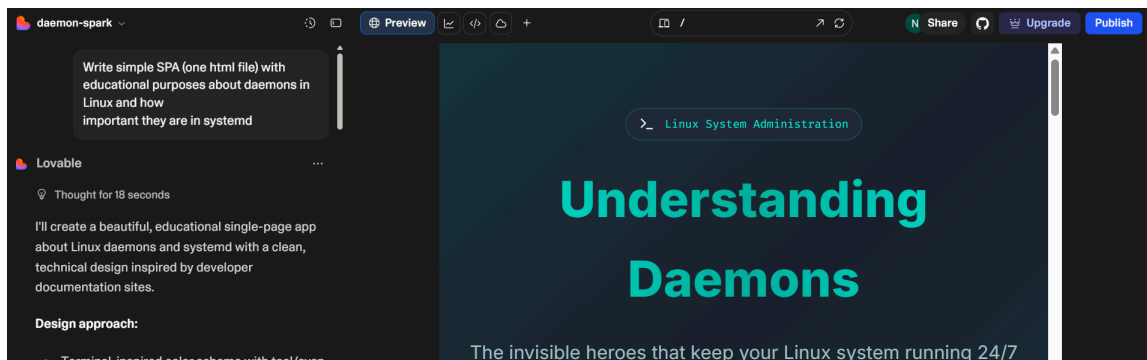
5. Next step for privacy is changing configuration of `/etc/systemd/system/secure-nginx.service`

```
➤ sudo chown -R webuser:webuser /var/www/html

(nikita@nikita)-[~]
$ sudo chown -R webuser:webuser /var/www/html

(nikita@nikita)-[~]
$ sudo chown -R webuser:webuser /var/log/nginx
```

6. Make readable `nginx` log and web site page for a user.



7. With Lovable AI I have created simple SPA to host on the web-server. I wrote the SPA inside `var/www/html/index.html`

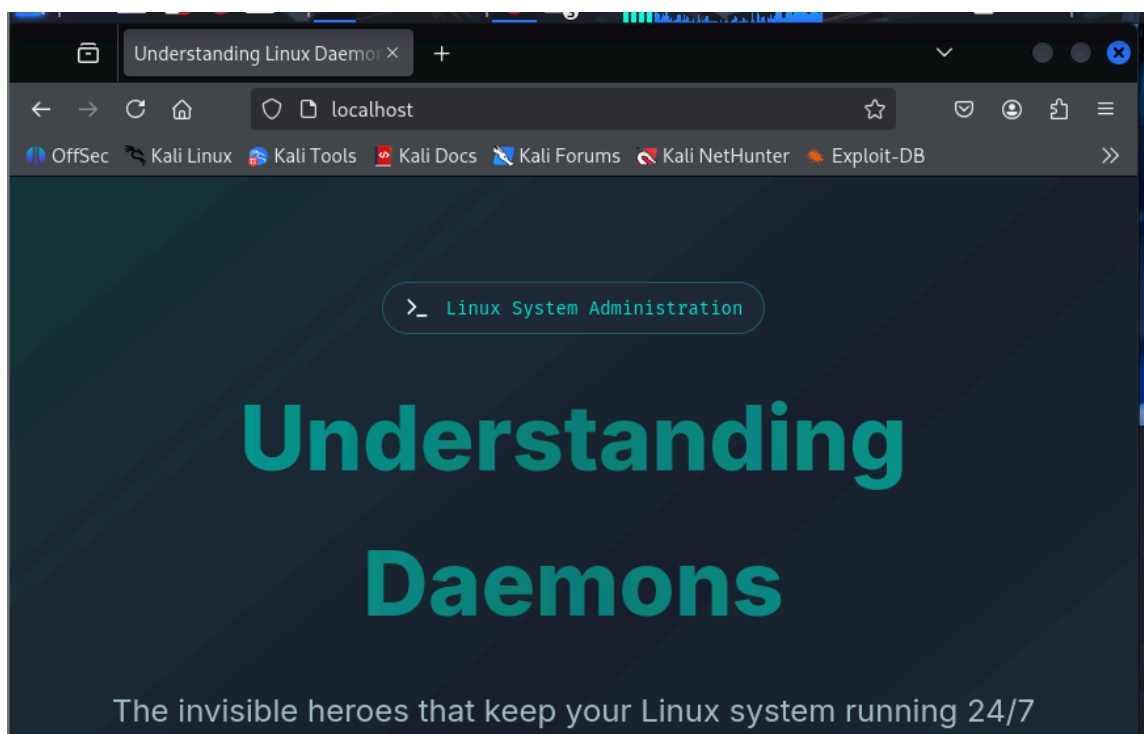
```
(nikita@nikita)-[~]
$ sudo chown -R webuser:webuser /var/log/nginx

(nikita@nikita)-[~]
$ sudo systemctl daemon-reload

(nikita@nikita)-[~]
$ sudo systemctl enable secure-nginx.service
Created symlink '/etc/systemd/system/multi-user.target.wants/secure-nginx.service' → '/etc/systemd/system/secure-nginx.service'.

(nikita@nikita)-[~]
$ sudo systemctl start secure-nginx.service
```

8. Then I refresh daemons and enable and start secure nginx service



```
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
Active: active (running) since Wed 2025-10-15 23:51:26 +03; 1min 40s ago
Invocation: 8e290773b51240d4b208920eb127fb98
Docs: man:nginx(8)
Process: 84435 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
Process: 84436 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
Main PID: 84438 (nginx)
Tasks: 5 (limit: 2206)
Memory: 5.1M (peak: 5.6M)
CPU: 40ms
CGroup: /system.slice/nginx.service
├─84438 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
├─84439 "nginx: worker process"
├─84440 "nginx: worker process"
├─84441 "nginx: worker process"
└─84442 "nginx: worker process"

Oct 15 23:51:26 nikita systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy s
Oct 15 23:51:26 nikita systemd[1]: Started nginx.service - A high performance web server and a reverse proxy se
```

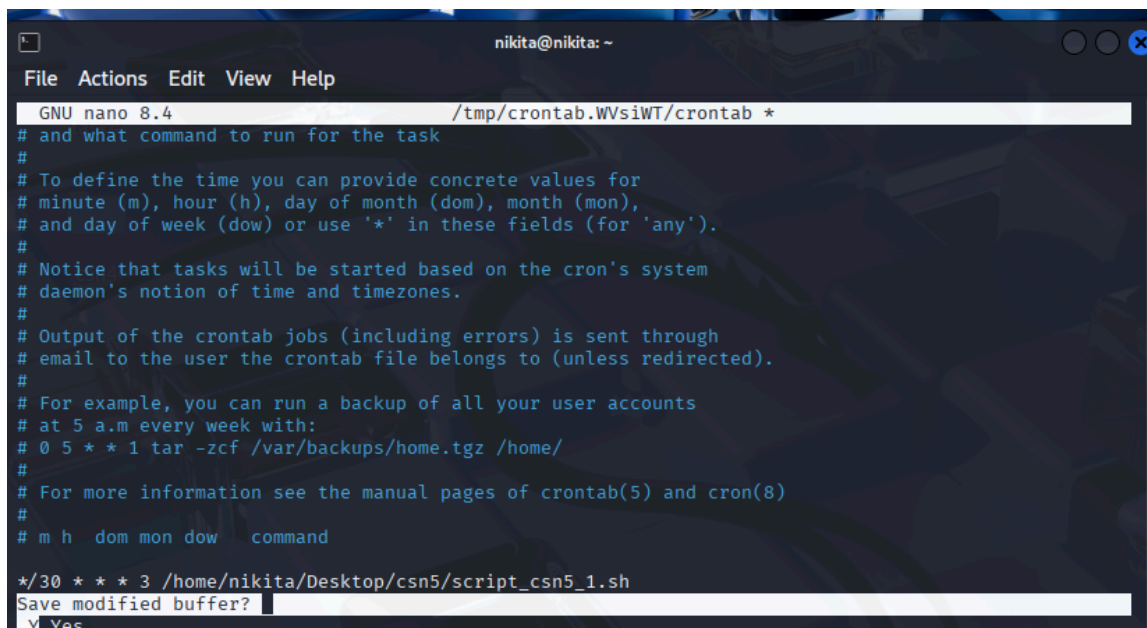
▼ Task 4: Crontab

Many server providers (hoster.by / timepad) allow convenient GUI to edit crontab. Also web servers written in Bitrix framework give abilities to edit it user friendly.

```
(nikita@nikita)-[~]
$ crontab -e
no crontab for nikita - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano          ← easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny

Choose 1-3 [1]: 1
crontab: installing new crontab
```

1. I execute crontab and set default view editor.



```
File Actions Edit View Help
GNU nano 8.4 /tmp/crontab.WVsiWT/crontab *
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
*/30 * * * 3 /home/nikita/Desktop/csn5/script_csn5_1.sh
Save modified buffer? [Y] Yes
```

2. I Delete all commentaries and set my crontab command

3. first * is for minutes I set every 30 minutes, others for default * and last is the day of week (3 stands for Wednesday) ; after that is the command that executes, in my case is a script

```
crontab: installing new crontab  
(nikita@nikita)-[~]  
$ crontab -l  
*/30 * * * 3 /home/nikita/Desktop/csn5/script_csn5_1.sh  
(nikita@nikita)-[~]
```

4. With -l flag I check and verify